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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester(2025-2026)
Faculty Name	Ms. G. Mamatha
Student Name	K. Sai Samyuktha
Roll Number	2520030443
Branch	CSE
Course Outcome	CO3-Breadth First Search ,Depth First Search , Minimum Spanning Tree

Case Study Topic: Hospital Department Connectivity and Communication Network Optimization using Graph Algorithms in HealthNet System

Work Done Summary:

In this case study, hospital departments are represented as a graph to analyze connectivity and optimize communication networks. BFS and DFS algorithms are used to traverse department networks and identify reachable departments, while the Minimum Spanning Tree (MST) algorithm is implemented to design a cost-effective communication network between hospital units.

Implementation Details:

1. Hospital departments are modeled as graph vertices.
2. Communication links between departments are represented as graph edges.
3. BFS (Breadth First Search) is used to find and analyze connectivity between departments.
4. DFS (Depth First Search) is used to traverse the hospital workflow network and detect connected components.
5. Minimum Spanning Tree (MST) is implemented to establish the minimum-cost communication network.
6. Adjacency list representation is used for efficient graph storage.

7. The system helps optimize communication, referrals, and resource sharing among departments.

Result : Screenshots / Output

```
Departments:
0 -> Emergency
1 -> ICU
2 -> Radiology
3 -> Pharmacy
4 -> Lab

Final MST (Hospital Communication Network)

Edge    Weight
0 - 1   2
1 - 2   3
0 - 3   6
1 - 4   5
```

Time Complexity :

1. Graph Representation: $O(V^2)$
2. Minimum Spanning Tree Construction: $O(V^2)$
3. Edge Selection Process: $O(V^2)$
4. Space Complexity: $O(V)$
5. The Minimum Spanning Tree (MST) efficiently connected all hospital departments with the minimum possible communication cost while maintaining full network connectivity.

GitHub Link: https://github.com/saisamyuktha471/DSA2-CO-3

Student Signature	**Faculty Signature**